Zhengjie (Jane) Yang

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Education

Stanford University 2024-2026
M.S., Civil and Environmental Engineering (Atmosphere/Energy)

University of Pennsylvania 2019-2021
M.S., Electrical and Systems Engineering

Chongqing University 2015-2019
B.Eng., New Energy Science and Engineering

Research Experience

Stanford University, Precourt Institute for Energy 2025-Present
Graduate Researcher | Data Center Energy Infrastructure

DC2: AC and DC power-delivery architectures

- Model and compare AC and DC power-delivery architectures for AI data centers, examining conversion losses, power quality, and energy-storage integration.
- Evaluate system-level design trade-offs and communicate findings through technical analysis and a CERAWeek 2026 research poster with Liang Min.

BRIDGES: Data-center demand and coupled energy-system planning

- Extend the BRIDGES energy-system model to examine how data-center demand affects electricity generation, natural-gas use, infrastructure investment, and system costs.
- Compare grid-connected and onsite power-supply configurations under consistent reliability requirements, using scenario analysis and numerical checks to assess cost and emissions trade-offs.

Chongqing University 2016-2019
Undergraduate Researcher | Biohydrogen Production | Advisor: Ao Xia

- Studied biohydrogen production using electrodialysis-based removal of volatile fatty acids and photovoltaic-thermal integration.
- Co-inventor of two patents on solar-coupled electrodialysis and organic-wastewater hydrogen-production systems, granted in 2020 and 2021.

Selected Publications

1. He, Z., Li, Z., **Yang, Z.**, Han, X., Wan, C., Ju, P., and Xu, Y. (2026). Analyzing the Aggregate Flexibility of Generalized Energy Storage Resources: An Enhanced Region Learning Method. *IEEE Transactions on Smart Grid*, 17(4), 3170-3181. doi:10.1109/TSG.2026.3655773.
2. Hua, H., Jiang, G., Chao, G., and **Yang, Z.** (2026). Can corporate green strategic actions enhance green innovation performance? *International Review of Financial Analysis*, 118, 105421. doi:10.1016/j.irfa.2026.105421.
3. Li, Y., **Yang, Z.**, and Dong, X. (2025). Green bond and greenwashing: New insights from Chinese firms. *Finance Research Letters*, 85, 107847. doi:10.1016/j.frl.2025.107847.

Selected Poster

Yang, J., and Min, L. (2026). *Direct Current (DC) Subtransmission Backbone for AI Factories.* CERAWeek 2026. Research poster.

Selected Academic Projects

Storage Planning for Curtailment and Congestion 2024
Stanford University | ENERGY 276 | Liang Min

- Modeled the WECC/CAISO system in PyPSA and Atlite to evaluate storage siting under renewable-generation expansion and transmission constraints.
- Compared demand-growth and zero-carbon scenarios, examining renewable curtailment, congestion, locational marginal prices, and storage deployment.

Relevant Industry Experience

Anima Power Corp. Jan 2026-Present
Consultant | AI Infrastructure and Energy Storage

- Evaluate data-center power and BESS opportunities, including site readiness, utility interconnection, onsite generation, and technology configuration.

GridCARE, Inc. Jul 2024-Sep 2025
Product Strategy Manager (Jul 2024-Apr 2025);
Business Development Manager (May-Sep 2025)

- Led early data-center customer development, working with large-load customers and utility teams to understand power requirements, site-selection needs, and expansion opportunities.
- Framed customer needs as technical questions for engineering assessment, including power availability, interconnection timing, and service-availability requirements.
- Worked with technical teams to clarify assumptions and communicate findings and trade-offs to customers; translated customer feedback into product priorities.

Qcells USA Nov 2021-Jul 2024
Associate Manager, Systems Integration Applications (Oct 2022-Jul 2024);
ESS Analyst, Project Development / Systems Integration Applications (Nov 2021-Sep 2022)

- Developed BESS sizing and techno-economic tools integrating battery/inverter configuration, degradation, augmentation, capital and operating costs, revenue, and project-finance assumptions.
- Built production-cost-modeling and machine-learning-assisted revenue-analysis workflows across ERCOT, CAISO, NYISO, and ISO-NE; evaluated more than 7 GW of early-stage BESS opportunities.
- Performed sizing, revenue modeling, and techno-economic evaluation for a six-project ERCOT portfolio totaling more than 1 GW / 2 GWh, including the 190 MW / 380 MWh Cunningham project.

Patents

Solar-Coupled Electrodialysis Biological Hydrogen Production Device and Method
Co-inventor. CN108504551B. Granted November 6, 2020.

Organic Wastewater Hydrogen Production System Based on a Photovoltaic-Thermal Collector
Co-inventor. CN108485952B. Granted March 16, 2021.

Technical Skills

Methods: Energy-system modeling; techno-economic analysis; BESS sizing; revenue modeling; scenario and sensitivity analysis; optimization.

Tools: Python, MATLAB, SQL, PyPSA, Atlite, CVXPY, Gurobi, Git, LaTeX.